Programs on Problem Solving

a: Solve the 8 Queens Problem

Aim:

To place 8 queens on an 8x8 chessboard so that no two queens attack each other, i.e., no two queens share the same row, column, or diagonal.

Procedure:

1. Create an 8x8 board initialized with 0s.
2. Place a queen in each row by checking if it’s safe.
3. If no column is safe in a row, backtrack.
4. Repeat until all 8 queens are placed.

Python Code:

```
N = 8
count=0

def safe(board, row, col):
    for i in range(row):
        if board[i] == col or abs(board[i]-col) == abs(i-row):
            return False
    return True

def solve(board, row):
    global count

    if row == N:
        count += 1
        print("Solution", count)
        for i in range(N):
            print(". "*board[i] + "Q " + ". "*(N-board[i]-1))
        print()
        return

    for col in range(N):
        if safe(board, row, col):
            board[row] = col
            solve(board, row+1)

board = [-1] * N
```

```
solve(board, 0)
print("Total Solutions:", count)
```

Output:

```
Solution 1
Q . . . . . . .
. . . . Q . . .
. . . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . . Q .
. Q . . . . . .
. . . Q . . . .

Solution 2
Q . . . . . . .
. . . . . Q . .
. . . . . . . Q
. . Q . . . . .
. . . . . . Q .
. . . Q . . . .
. Q . . . . . .
. . . . Q . . .

Solution 3
Q . . . . . . .
. . . . . . Q .
. . . Q . . . .
. . . . . Q . .
. . . . . . . Q
. Q . . . . . .
. . . . Q . . .
. . Q . . . . .

Solution 4
Q . . . . . . .
. . . . . . Q .
. . . . Q . . .
. . . . . . . Q
. Q . . . . . .
. . . Q . . . .
. . . . . Q . .
```

. . Q . . . . .

Result:

Thus the program of placing 8 queens on an 8x8 chessboard was executed successfully.